

Fetal Health Classification

Machine Learning — Evaluation Project

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Motivation

- **Cardiotocography (CTG)** records fetal heart rate and uterine contractions during pregnancy; the standard, low-cost, non-invasive tool obstetricians use to monitor fetal wellbeing in real time.
- **Problem:** correctly separating Normal from Suspect and Pathological patterns matters clinically; missed distress signals can lead to preventable stillbirth or birth injury, while false alarms drive unnecessary interventions such as emergency C-sections.
- CTG traces are still interpreted manually today, and inter-clinician agreement is known to be limited; creating a natural opening for a consistent, data-driven decision-support classifier.
- **Goal:** given 21 diagnostic features automatically extracted from a CTG recording, predict whether the fetal state is Normal, Suspect, or Pathological.
- **Challenge:** the dataset is imbalanced; most recordings are Normal, making standard accuracy a misleading metric.



Data - Overview

Source: UCI / Kaggle Fetal Health Dataset

Samples: 2,126 CTG recordings

Features: 21 (signal + histogram statistics)

Classes: 3

Missing values: 0

Number of duplicate rows: 13

Feature groups:

- FHR baseline, accelerations, fetal movement, uterine contractions
- Decelerations: light / severe / prolonged
- Short- & long-term variability measures
- FHR histogram statistics (width, min/max, mode, mean, median, variance...)

Target feature → fetal_health

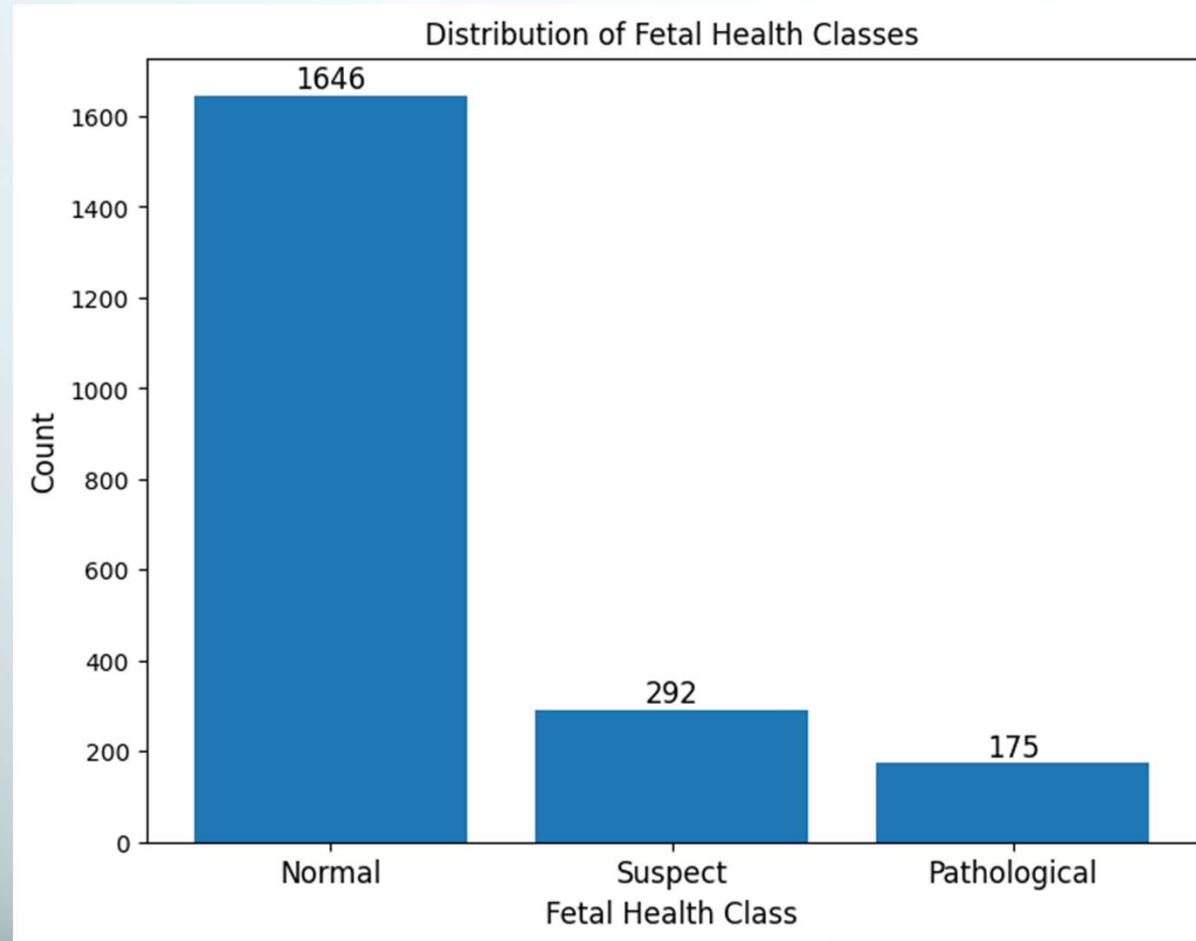
- 1 → Normal
- 2 → Suspect
- 3 → Pathological

Data - Sample

baseline value	accelerations	prolongued_ decelerations	abnormal_ short_term_ _variability	mean_value_of_ short_term_varia _bility	histogram_ mean	fetal_ health
120.0	0.000	0.0	73.0	0.5	137.0	2.0
132.0	0.006	0.0	17.0	2.1	136.0	1.0
133.0	0.003	0.0	16.0	2.1	135.0	1.0
134.0	0.003	0.0	16.0	2.4	134.0	1.0
134.0	0.001	0.002	26.0	5.9	107.0	3.0

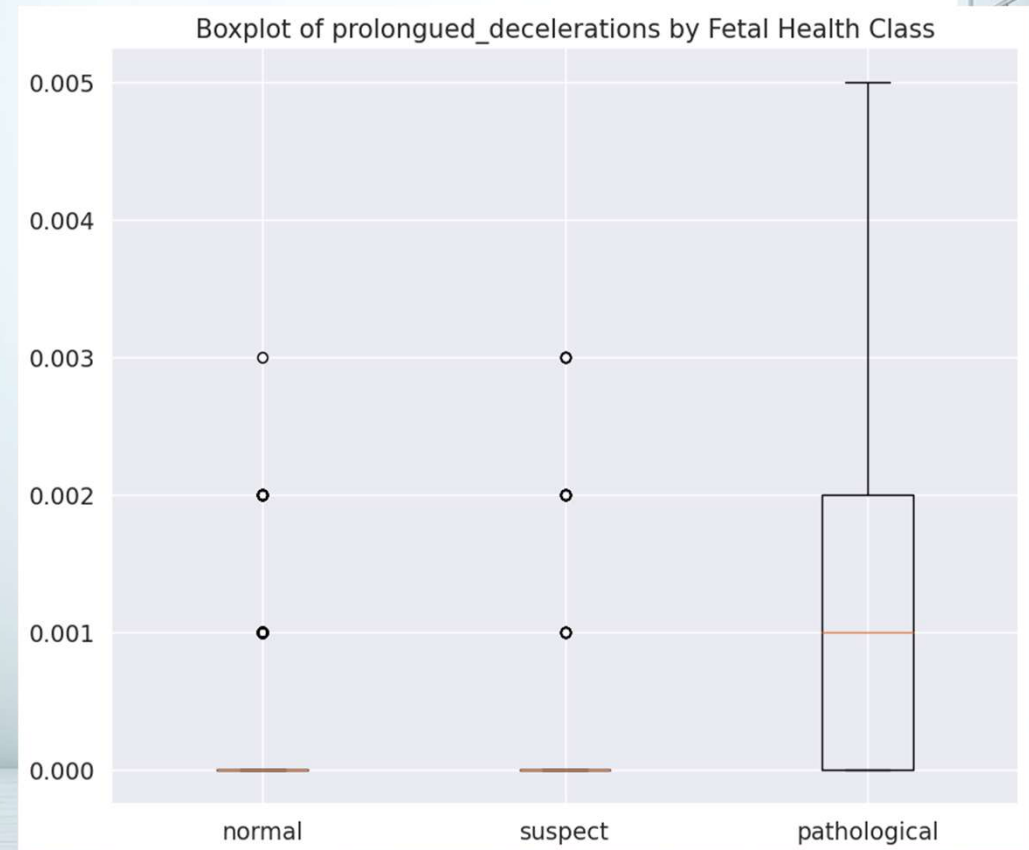
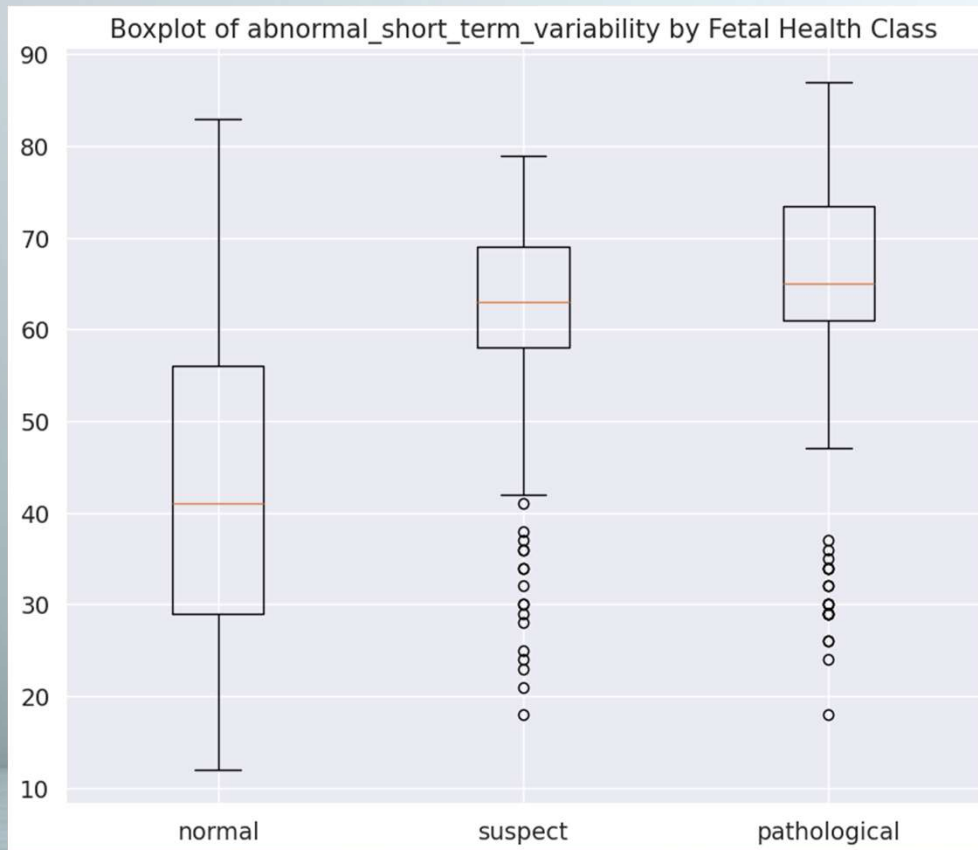
Data – Class Distribution

- The dataset is significantly imbalanced. Most recordings are Normal.
- Directly motivates macro-F1 as the evaluation metric. Accuracy alone is misleading.
- A stratified train/test split ensures both sets preserve the original class proportions.
- Oversampling minority classes was assessed to counteract the heavy dominant class bias.



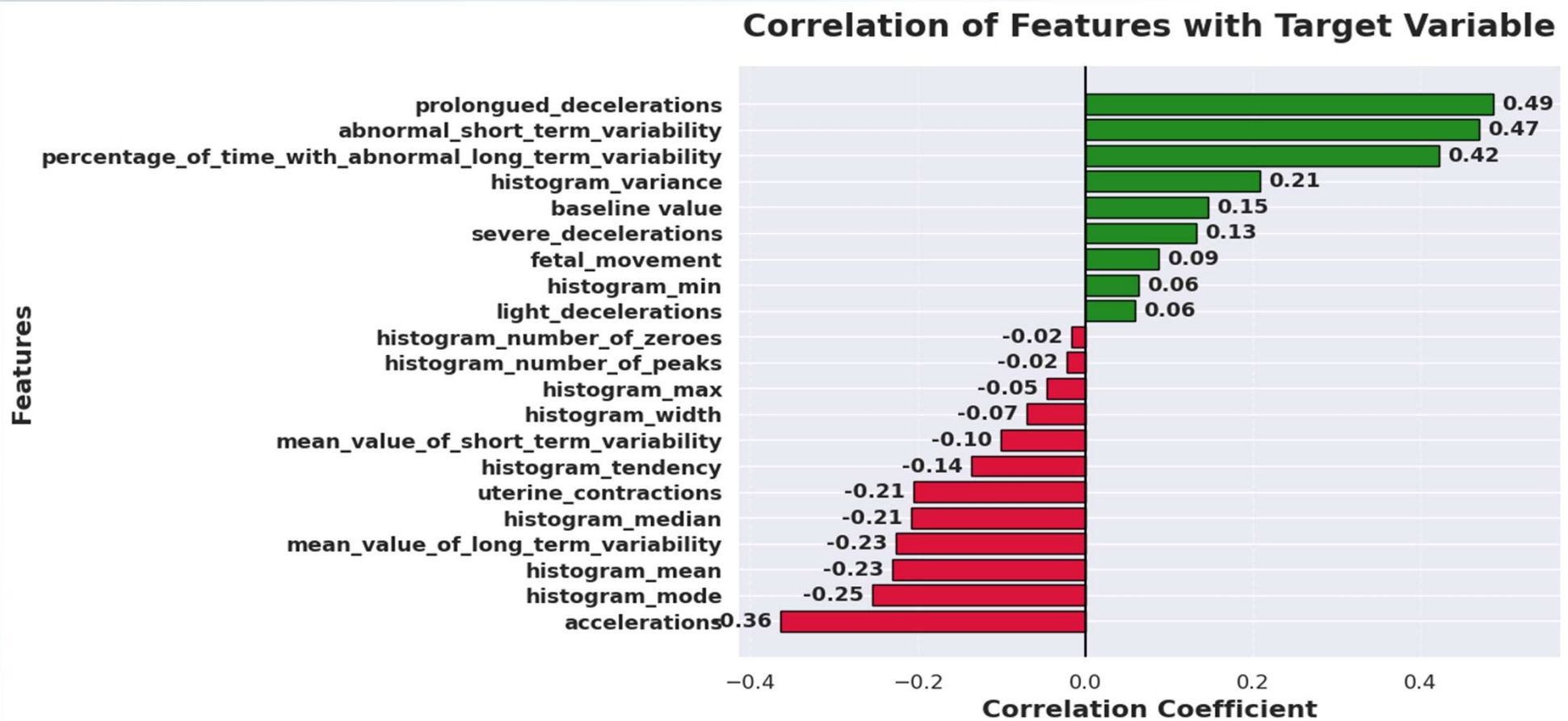
Data – Feature Exploration

- Pathological cases are clearly separated in most features.
- The Suspect class overlaps heavily with Normal in several features; The main source of classification difficulty.

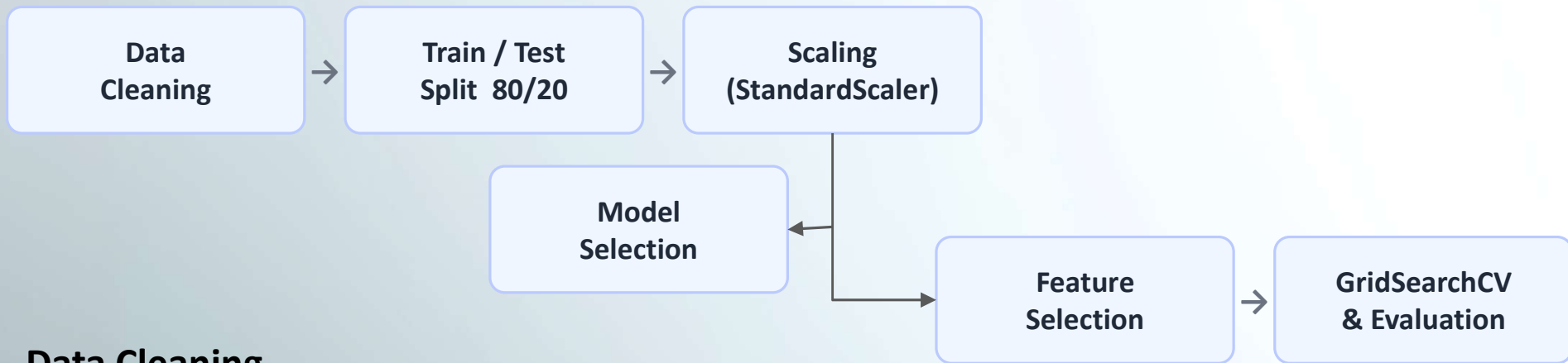


Data – Feature-Target Correlations

several features have low correlation → motivates feature selection



Method – Pipeline Architecture

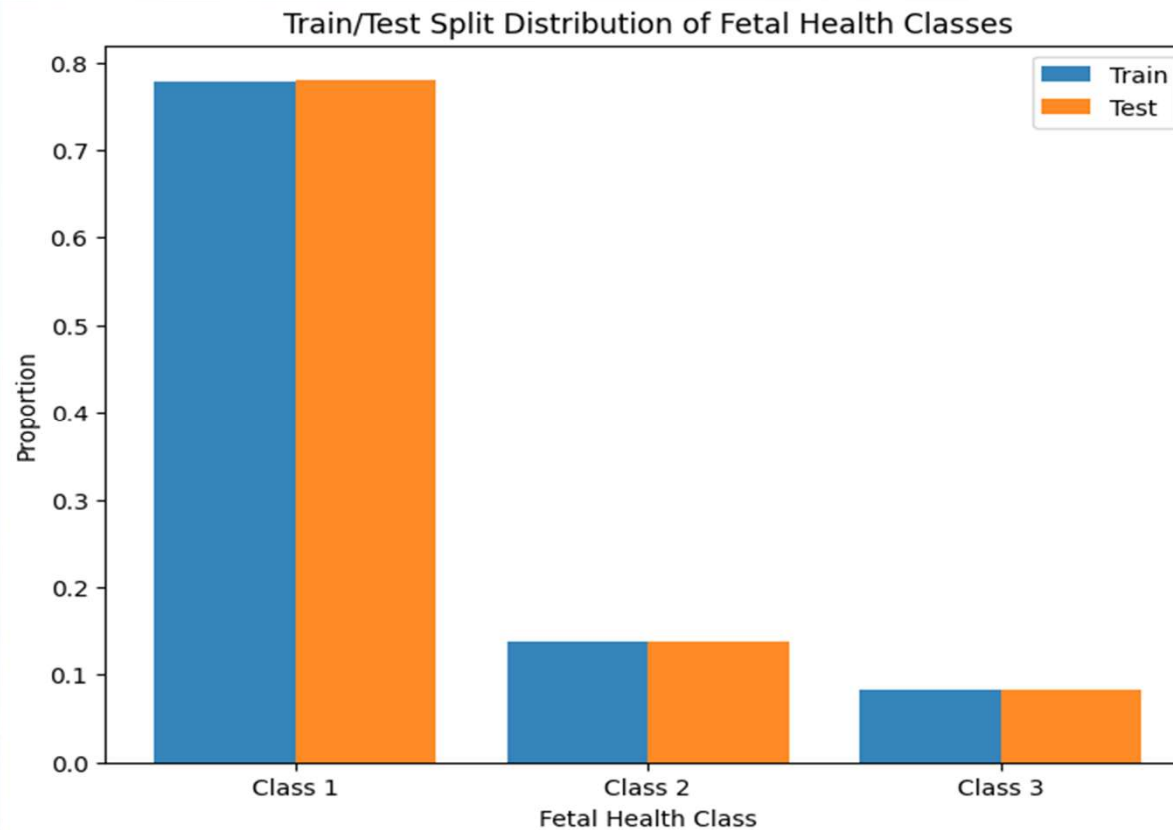


Data Cleaning

- 13 duplicates removed (Now 2113 samples).
- No features with missing values.
- No removal of outliers

Method – Stratified Train/Test Split

- Split ratio: 80% training (1690 samples) · 20% test (423 samples).
- Stratified by target class. Class proportions are preserved in both sets.
- Test set is touched exactly once which is after all tuning is complete.



Method – Classifiers Selected

Base Classifiers

KNN:

Tunned Hyperparameter: n_neighbours

Logistic Regression:

Tunned Hyperparameter: C (Regularization)

Advanced Classifiers

Support Vector Classifier (SVC):

Tunned Hyperparameters:

- C → [0.1, 1, 10, 100]
- Gamma → [“Scale”, “auto”, 0.01, 1]
- Kernel → [“rbf”, “linear”, “poly”, “sigmoid”]

Random Forest Classifier:

Tunned Hyperparameters:

- n_estimators → [100, 200, 300]
- max_depth → [None, 5, 10, 20]
- min_sample_split → [2, 5, 10]
- max_features → [“sqrt”, “log2”]

Method – Feature Selection

- Sequential Feature Selector (SFS), forward direction. Adds the single best feature at each step.
- Fixed classifier inside SFS (fixed params). Avoids the cost of nested hyperparameter tuning.
- Grid searches over `n_features_to_select`: [5, 10, 15, 20].
- RFC uses RFECV instead which leverages the tree's built-in feature importance scores directly.

Method – Cross-Validation & Hyperparameter Tunning

- GridSearchCV with StratifiedKFold (5-fold outer CV).
- SFS uses 3-fold inner CV. Kept lower to limit computation time.
- **Scoring metric:** macro-F1. Treats all three classes equally regardless of size.
- **Results:** CV mean and std on training data, final score on held-out test set.

Why macro-F1?

With 1,646 Normal vs 175 Pathological samples, accuracy is misleading. Macro-F1 treats all three classes equally.

Handling Class Imbalance

Both **SMOTE (Synthetic Minority Oversampling Technique)** and **class_weight="balanced"** were evaluated. SMOTE consistently outperformed class weighting across all models and was therefore selected as the sole imbalance handling strategy and applied inside each CV pipeline on training folds only.

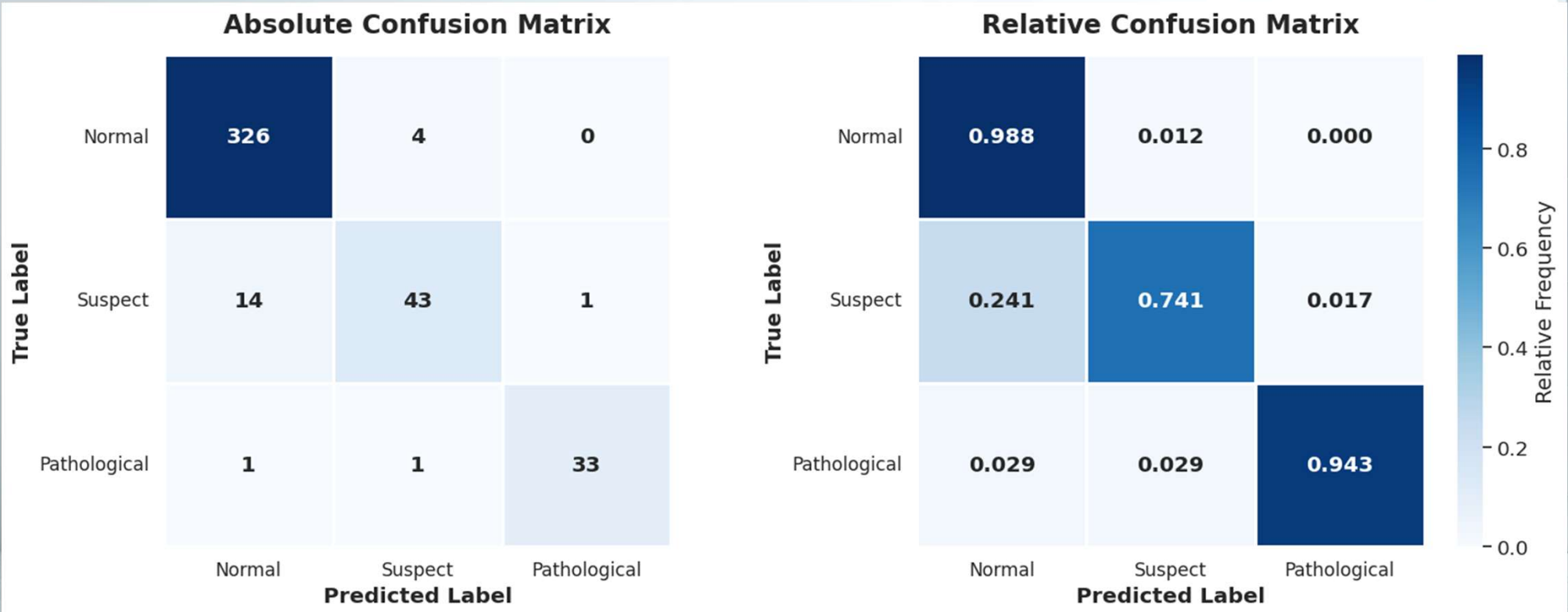
Results – Model Comparison

All models trained with the same pipeline: scaling → feature selection → GridSearchCV (5-fold, macro-F1).

Model	Best Params	Selected Features	CV F1 (mean, std)	Test F1 (macro)	Test Accuracy
KNN					
Logistic Regression					
SVC					
RFC					

Results – Best Model

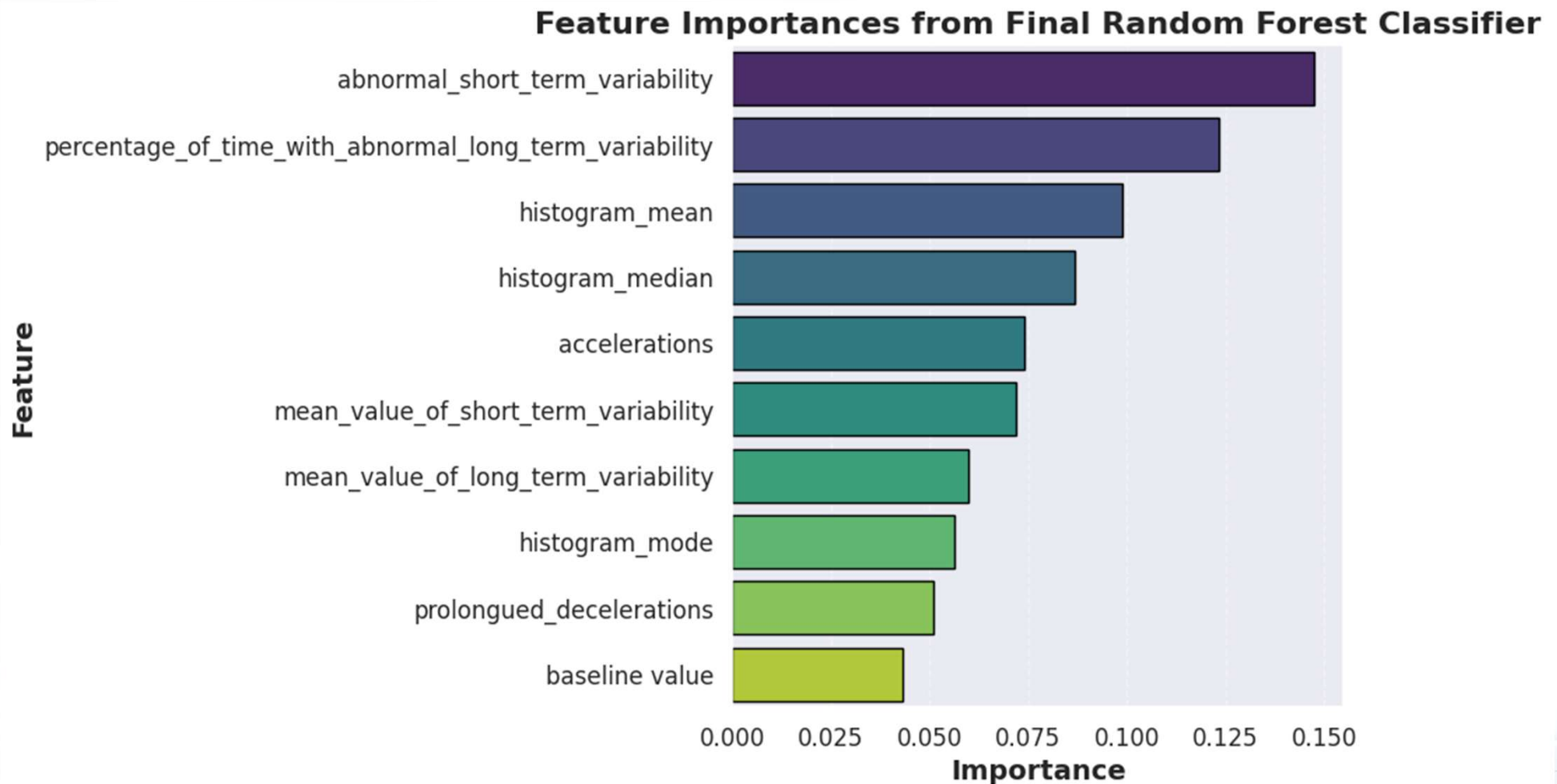
- Confusion matrix of RFC on the held-out test set.
- **Key observation:** Most errors involve the Suspect class — confused with both Normal and Pathological.



Results – Why RFC outperformed the others

- **Non-linear boundaries:** The Normal/Suspect boundary is non-linear; RFC models it naturally through recursive splits without shape assumptions.
- **Feature interactions:** RFC captures combinations feature at every split; LogReg treats each feature independently.
- **Ensemble stability:** Averaging across hundreds of trees cancels individual errors.
- **No distributional assumptions:** Skewed, non-Gaussian features do not affect tree splits; LogReg and SVC implicitly expect well-behaved distributions.
- **Outlier robustness:** Splits are threshold-based, not distance-based, so retained outliers do not distort RFC the way they distort KNN and SVC.

Results – Why RFC outperformed the others



Discussion

- Advanced classifiers (RFC, SVC) outperformed base classifiers across all metrics confirming that the Normal/Suspect boundary requires non-linear modelling.
- The Suspect class remained the hardest across all four models.
- KNN overfitting (train F1 = 1.0, val F1 $\approx$ 0.6) at k=1 was the clearest example of how hyperparameter tuning directly impacts generalisation.
- Placing SMOTE inside the CV pipeline was critical; applying it before splitting would have inflated CV scores by leaking synthetic samples into the validation folds.

Discussion

- The learning curve had not plateaued; more real data could further improve minority class performance, particularly for Pathological.

Summary

- Dataset of 2,126 CTG recordings, 21 features, 3 imbalanced classes; macro-F1 chosen as the evaluation metric.
- Data cleaning: 13 duplicate rows removed, outliers retained as potential pathological indicators.
- Feature selection reduced 21 features to [N] most informative; SFS for KNN, LogReg, SVC; RFECV for RFC.
- Advanced classifiers (RFC, SVC) outperformed base classifiers (KNN, LogReg) across all metrics.
- Best model: RFC ; Test F1 = [X.XXX] · CV F1 = [X.XXX ± X.XXX].
- Main challenge: Suspect class; highest inter-class confusion across all models.

The background of the slide is a bright, minimalist room. On the right side, there is a large window with a grid pattern, through which bright light is streaming, creating a strong lens flare effect across the scene. The floor is made of light-colored square tiles. The overall atmosphere is clean and modern.

Thank You – Questions?